Curriculum Vitae

Lukas Niebel

Postdoctoral researcher, University of Münster

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Education

04/2019–06/2023 **PhD Mathematics**, *Ulm University*
Thesis: Analytic aspects of kinetic partial differential equations
Advisor: Prof. Dr Rico Zacher
Referees: Prof. Dr Cyril Imbert, Prof. Dr Hongjie Dong
Grade: Summa cum laude; awarded by the Ulmer Universitätsgesellschaft

03/2018–03/2019 **MSc Mathematics**, *Ulm University*
Thesis: Kolmogorov equations – Well-posedness, regularity, asymptotics and Harnack inequalities
Grade: 1.0 with distinction; awarded for the best graduation

10/2014–02/2018 **BSc Mathematics**, *Ulm University*
Thesis: Long-time behaviour of Markov chains by discrete functional inequalities and entropic Ricci curvature
Grade: 1.1 with distinction; awarded for the best graduation

04/2013–09/2014 **Junior studies in mathematics**, *Fernuniversität Hagen and PFH Göttingen*

06/2014 **Abitur**, *Joachim–Hahn–Gymnasium Blaubeuren*

Awards and Scholarships

02/2025 PhD thesis award "Promotionspreis der Ulmer Universitätsgesellschaft" (1500 EUR)

06/2022 Travel grant by the ProTrainU Mobility-Programme, Ulm University (1000 EUR)

04/2022 Master graduation award "Absolventenpreis M.Sc. Mathematik Uni Ulm 2020" for the best graduation (500 EUR)

11/2019 Graduate scholarship granted by the State of Baden-Wuerttemberg, Germany (52,200 EUR)

05/2018 Bachelor graduation award "Absolventenpreis B.Sc. Mathematik Uni Ulm 2018" for the best graduation (500 EUR)

06/2014 Ferry Porsche award and graduation award of the Deutsche Physikalische Gesellschaft

Research

I develop geometric and analytic tools for kinetic and hypoelliptic partial differential equations and for free-boundary fluid dynamics.

Preprints

[1] **Kinetic Sobolev Spaces** with Pascal Auscher. 2026. arXiv:2603.17491.

[2] **Nash's G bound for the Kolmogorov equation** with Helge Dietert. arXiv:2510.21621, 2025.

[3] **Global rigidity of two-dimensional bubbles**. arXiv:2510.17557, 2025.

[4] **Critical trajectories in kinetic geometry** with Helge Dietert, Clément Mouhot, and Rico Zacher. arXiv:2508.14868, 2025.

[5] **Weak solutions to Kolmogorov–Fokker–Planck equations: regularity, existence and uniqueness** with Pascal Auscher and Cyril Imbert. arXiv:2403.17464, 2024.

Publications

[6] **Steady bubbles and drops in inviscid fluids** with David Meyer and Christian Seis. *Calc. Var.*, 64(299), 2025.

[7] **A trajectorial interpretation of Moser's proof of the Harnack inequality** with Rico Zacher. *Ann. Sc. Norm. Super. Pisa Cl. Sci. (5)*, 26(2):605–617, 2025.

[8] **Fundamental solutions to Kolmogorov–Fokker–Planck equations with rough coefficients: existence, uniqueness, upper estimates** with Pascal Auscher and Cyril Imbert. *SIAM J. Math. Anal.*, 57(2):2114–2137, 2025.

[9] **On a kinetic Poincaré inequality and beyond** with Rico Zacher. *J. Funct. Anal.*, 289(1):Paper No. 110899, 18, 2025.

[10] **Analytic aspects of kinetic partial differential equations**. *OPARU — Open Access Repositorium der Universität Ulm*, 2023.

[11] **Kinetic maximal L^p-regularity with temporal weights and application to quasilinear kinetic diffusion equations** with Rico Zacher. *J. Differential Equations*, 307:29–82, 2022.

[12] **Kinetic maximal $L^p_\mu(L^p)$-regularity for the fractional Kolmogorov equation with variable density**. *Nonlinear Anal.*, 214:Paper No. 112517, 21, 2022.

[13] **Kinetic maximal L^2-regularity for the (fractional) Kolmogorov equation** with Rico Zacher. *J. Evol. Equ.*, 21(3):3585–3612, 2021.

Employment History

04/2023–06/2026 **Postdoctoral researcher in the group of Prof. Dr Christian Seis**, *University of Münster*

10/2022–03/2023 **Predoctoral researcher**, *Ulm University*

03/2020–10/2022 **Teaching assistant**, *University of Applied Sciences Neu-Ulm*

10/2015–10/2019 **Teaching assistant**, *Ulm University*

03/2015–03/2019 **Research assistant (working student)**, *Daimler AG Research & Development Ulm*

Teaching Experience

University of Münster

2023–2025 Teaching assistant for *Partial Differential Equations* (1 sem.) and *Analysis and Numerics of Differential Equations* (1 sem.) (exercise-sheet design; large-class solution sessions; tutor and marking coordination; exam support)

Winter 2024 Supervised a student seminar on *De Giorgi–Nash–Moser Theory*

2024–2025 Examined two BSc theses; co-supervised and examined the MSc thesis of Yuanjiang Han (now a PhD student with Christian Seis)

University of Applied Sciences Neu-Ulm

2020–2022 Teaching assistant for *Business Mathematics and Statistics I* (3 sem.) and *Business Mathematics and Statistics II – Regression Models / Statistical Testing* (2 sem.) (authored exercise sheets; ran the programming lab; marked submissions; created teaching videos)

Ulm University

2017–2022 Teaching assistant for *Analysis I* (2 sem.), *Analysis II* (1 sem.), *Dynamical Systems* (2 sem.), and *Elements of Functional Analysis* (1 sem.) (exercise-sheet design; large-class solution sessions; tutor and marking coordination; exam-setting support)

Winter 2022 Local coordinator of the 26th Internetseminar

Winter 2019 Organised the "Trainingscamp", an annual four-week maths skills refresher for first-year students

2015–2017 Mathlab tutor for *Analysis I* and *Analysis II*

Talks and Posters

03/2026 Talk – Horizons in nonlinear PDEs, Ulm University

10/2025 Talk – Oberseminar Angewandte Analysis, Dortmund University

10/2025 Talk – Noko25 43rd Northern German Colloquium, Humboldt-Universität zu Berlin

09/2025 Poster – Hyperbolic Problems in Nürnberg 2025, FAU Erlangen

05/2025 Talk – Seminar Nichtlineare Partielle Differentialgleichungen, KIT Karlsruhe

03/2025 Poster – Kinetic equations and turbulence, CentraleSupélec and Collège de France

10/2024 Talk – Münster–Imperial Day in PDE, University of Münster

09/2024 Talk – Oberseminar Analysis, University of Bonn

03/2024 Poster – Mathematics Münster Mid-term Conference, University of Münster

11/2023 Talk – Séminaire Analyse Numérique et EDP, Université Paris-Saclay

09/2023 Talk – Seminar of the AG Analysis-Probability, MPI Leipzig

06/2023 Talk – Real Analysis and Geometry, CIRM Marseille

04/2023 Talk – Wilhelm Killing Colloquium, University of Münster
03/2023 Talk – Chair Dynamics Control and Numerics, FAU Erlangen
02/2023 Talk – TULKKA, KIT Karlsruhe
11/2022 Poster – Kinetic Theory, CIRM Marseille
09/2022 Talk & Poster – Horizons in non-linear PDEs, Ulm University
07/2022 Poster – When Kinetic Theory meets Fluid Mechanics, ETH Zürich
06/2022 Poster – 11th Summer school on "Methods and Models of Kinetic Theory", Pesaro
01/2022 Poster – Frontiers in analysis of kinetic equations, INI Cambridge
07/2021 Talk – Nonlinear Evolution Equations and Approximations, University of Duisburg-Essen
03/2021 Talk – Oberseminar Analysis, Ulm University
02/2021 Poster – Winterschool on Analysis and Applied Mathematics, University of Münster

Research Stays

05/2026 Visit to Luis Silvestre (University of Chicago), one week
04/2026 Visit to Sam Punshon-Smith (Tulane University), one week
04/2026 Visit to David Herzog (Iowa State University), two weeks
2023–2025 Multiple visits to Pascal Auscher (Université Paris-Saclay), totalling 7 weeks
2023–2025 Multiple visits to Rico Zacher (Ulm University), totalling 5 weeks
2021–2025 Three visits to Cyril Imbert (École normale supérieure, Paris), totalling two weeks
05/2025 Visit to Björn de Rijk (KIT Karlsruhe), one week
09/2023 Visit to Tobias Ried (MPI Leipzig), one week
03/2023 Visit to Nicola De Nitti (FAU Erlangen), one week
02/2023 Visit to Christian Seis (University of Münster), half a week

Other Scientific Activities

2026 Supporting the organisation of the Münster–Imperial Day in PDE, University of Münster
2024 Supporting the organisation of the Münster–Imperial Day in PDE, University of Münster
2022 Supporting the organisation of the summer school "Horizons in non-linear PDEs" at Ulm University
2020 Co-organisation of the seminar "Mathematisches Kolloquium" at Ulm University for six semesters
2022 Organisation of a weekend workshop for mathematics PhD students at Ulm University: 15 participants, funding (2500 EUR) acquired by the StuVe Ulm University
2022 Supervision of a three-month project for the workshop of the 25th Internetseminar. Title: Kolmogorov equation in $L^p(\mathbb{R}^{2n})$ – Semigroup, spectrum, maximal regularity?
2021 Facilitator at the autumn school "COLLAB – Collaboratory for Global Sustainability 2021", Toronto/Ulm
2019 Supporting the organisation of the winter school "Grad. Flows and Var. Methods in PDEs" at Ulm University

Active contribution to the PDE seminars at the University of Münster and Ulm University by inviting several external speakers

Attended over 30 international workshops and conferences, including events at CIRM Marseille, INI Cambridge, ETH Zürich, and the University of Oxford

Refereeing activity for *Annales scientifiques de l'École normale supérieure*, *Communications in Partial Differential Equations*, *Discrete and Continuous Dynamical Systems*, *European Journal of Mathematics*, *Journal of Evolution Equations*, and *Potential Analysis*

Academic Volunteer Work

02/2021–09/2022 **Elected member of the council of PhD students**, *Ulm University*
Representing the interests of doctoral students in university committees, e.g. for the change of the PhD regulations; chair of the council of PhD students (12/2021–09/2022)

02/2021–09/2022 **Elected member of the faculty council for Mathematics and Economic Sciences**, *Ulm University*

07/2017–03/2019 **Member of the Student Council Mathematics**, *Ulm University*
Member of the appointment committee for a professorship in "Stochastics"